

1 Introduction

The field of international trade theory has evolved significantly over the past few centuries, beginning with the foundational concepts of absolute and comparative advantage proposed by Adam Smith and David Ricardo. These early models laid the groundwork for understanding the benefits of trade between nations with differing productivity levels. The Heckscher-Ohlin (H-O) model [1], developed in the early 20th century, extended these ideas by incorporating factor endowments, positing that countries export goods that utilize their abundant resources most intensively. Differences in the abundance of factor endowments, such as labour and capital, of different countries were used to study the patterns of production and subsequent trade flow between countries. It was attractive because it provided an elegant explanation for international trade flows based on measurable economic variables, such as the relative abundance of capital and labour, rather than relying on relatively abstract variables such as technology or productivity [2, 3].

The H-O model had faced significant empirical challenges in explaining international trade flows [4, 5]. The most notable challenge came from Wassily Leontief's empirical study in 1953, which showed that the United States, a country abundant in capital, was exporting labour-intensive goods and importing capital-intensive goods, contrary to the predictions of the H-O model [6]. Such discrepancies have prompted economists to refine and expand upon traditional trade models to better align theory with observed trade patterns [7]. The New Trade Theory introduced by Paul Krugman in the late 20th century emphasised the roles of economies of scale and product differentiation in shaping international trade [2, 4]. Unlike the H-O model, which relies on factor endowments, the New Trade Theory explains why countries with similar factor endowments engage in substantial trade by highlighting the benefits of economies of scale and the demand for a variety of goods [8, 9]. Krugman's model captures the empirical observation of intra-industry trade, where countries both export and import similar types of goods, something the H-O model fails to explain. Building on this, the Melitz model introduced firm-level heterogeneity, accounting for differences in productivity among firms within an industry [10]. Melitz's framework demonstrates how only the most productive firms can bear the fixed costs of exporting, leading to reallocating resources towards more efficient producers as trade liberalises. This model aligns well with empirical data showing that trade liberalisation increases industry productivity and firm-level turnover, with less productive firms exiting the market and more productive firms expanding their export activities. See [2] for a more detailed review. Despite the empirical challenges faced by the H-O model, its conceptual foundation remains influential. The factor-endowment framework is still relevant in explaining aspects of global production and trade, particularly when combined with mechanisms that account for economies of scale, firm heterogeneity, and capital mobility.

To overcome the challenges of the traditional trade models, agent-based models are particularly well-suited because they simulate macroeconomic outcomes by modelling individual decision-making. Unlike traditional equilibrium-based models, ABMs allow for emergent economic behavior, where complex trade patterns arise from simple rules governing labour and investment choices. While there are numerous ABMs designed

to study various aspects of the economy [11–16], relatively few focus specifically on international trade [17–19].

In this paper, we construct an agent-based modelling (ABM) framework to study international trade and development, based on the work of [20], which introduced an agent-based implementation of the H-O model in a simple yet novel way. While the model in [20] is highly stylised, designed to reproduce a few theoretical results from trade theory using an agent-based approach, it offers the flexibility to relax core assumptions and incorporate additional economic mechanisms crucial for a more accurate representation of global trade dynamics. We expand on the model by (i) scaling the model to simulate scenarios with n countries and m sectors, (ii) introducing additional features for the model such as tariffs, and (iii) building a method to calibrate the model to real-world trade flows data.

The remainder of the paper is structured as follows. Section 2 introduces the generalised agent-based model, detailing the mechanics of the model. Section 3 validates the model through theoretical case studies, while relaxing the standard assumptions and considering scenarios under increasing returns to scale, capital mobility and explore scenarios where tariffs are introduced. Section 4 focuses on the empirical calibration of the model using Multi-Region Input-Output data for a 3-country, 3-sector system with USA, Mexico, and China, highlighting the model’s ability to reproduce real-world trade patterns. Section 5 discusses the broader implications of the results, particularly under policy interventions like tariffs. Finally, Section 6 concludes by reflecting on the strengths and limitations of the framework and suggesting directions for future research.

2 Model

2.1 The Generalised Trade ABM

We provide a general overview of a generalised model proposed initially by [20] below. The model has two types of agents: a citizen and a nation. Citizens in each country make choices regarding their jobs and investments based on a stochastic process. The aggregated labour and capital of the citizens are used as inputs to production and consequently trade. Fig 1 provides a visual description of the flow of events in the model. Below are the steps followed by the model:

1. Citizens earn income from labour and investment choices with a probability, p_{reassess} reflecting the opportunity of switching jobs and investment choices to the one with the highest wage and return as possible.
2. Nations aggregate the labour and capital from their citizens.
3. Nations utilize labour and capital to produce goods.
4. Nations distribute wages and returns to citizens.
5. Nations may trade goods by purchasing goods from other nations in return for the nominal good.
6. Nations adjust prices of goods based on the demand and supply
7. Citizens consume goods based on their income and prices of goods.

Traditional comparative advantage models operate under the assumptions of (i) constant or decreasing returns to scale in all industries and (ii) internationally immobile capital. When these assumptions are relaxed, several complex economic phenomena emerge. Allowing for increasing returns to scale introduces the possibility of multiple equilibria, where small initial advantages can lead to long-term dominance by specific countries. Similarly, capital mobility complicates trade predictions, as it allows capital to flow toward high-return sectors across borders, making it more difficult to determine an optimal trade policy. In [20], the author constructs a simplified global economy with two countries and two sectors, similar to the framework developed by Samuelson in [21], to analyze the effects of outsourcing, increasing returns to scale [22], and capital mobility [23].

Below, we have a detailed exploration of the construction of the model, following each step outlined in Figure 1. While we present the analytical formulations governing the model dynamics, it is important to emphasize that this framework is fundamentally built for numerical simulations. The model operates as an agent-based system, where individual-level decisions collectively shape macroeconomic outcomes, making it well-suited for computational analysis rather than closed-form solutions.

Furthermore, the model is designed to be modular, allowing for flexibility in implementation and adaptation to different economic environments. While we describe a basic version of the model below, each component—including labor allocation, production functions, trade mechanisms, and pricing dynamics—can be replaced or modified with alternative functional forms or more sophisticated representations. This modularity ensures that the model can accommodate extensions, such as alternative production functions, adaptive trading, wage-setting mechanisms, and demand and consumption formulations, making it a versatile tool.

2.2 Labor and Investment choices

Each citizen i in country c is endowed with one unit of labor and one unit of capital, which they allocate to a production sector f and an investment sector f' respectively. These allocations are denoted as l_i : the sector in which citizen i is employed (labor choice), and k_i : the sector where citizen i invests their capital (investment choice).

Given these choices, the total income of an individual is defined as:

$$Y_i = W(c, f) + R(c, f') \quad (1)$$

Each citizen is characterized by the job and investment choices that he/she makes. The wage and returns from the sector f and f' are attained from the nation-level calculations shown below. At each time step, a citizen decides whether to reassess their labor and investment choices based on the probability p_{reassess} which governs the likelihood of switching sectors. If reassessment occurs, the individual chooses the sector offering the highest wage for labor allocation and the sector offering the highest return for capital allocation.

$$l_i^* = \arg \max_f W(c, f) \quad (2)$$

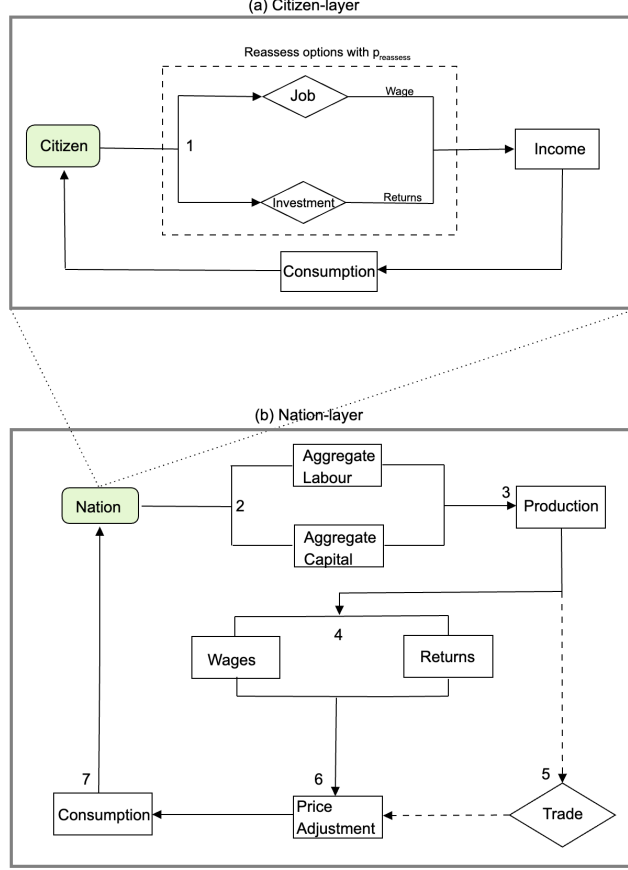


Fig. 1: Schematic of the agent-based model: The two figures provide a visual description of the sequence of events in the model. The top panel (a) represents the citizen-layer which showcases the labor, investment, and consumption decisions taken by each citizen in a country. The box with dotted lines represents the process of job and investment re-evaluation which is a stochastic process with the probability $p_{reassess}$. The bottom panel (b) gives the sequence of events in the nation-layer that occur following the events in the citizen-layer. This represents the authors' generalised reconstruction of the model presented in [20]

For investment reassessment, an agent reallocates capital to the sector with the highest return:

$$k_i^* = \arg \max_{f'} R(c, f') \quad (3)$$

where l_i^* , k_i^* represent the new sector choices after reassessment.

This highlights a bottom-up approach in the agent-based model. The behaviour of agents are governed by a very simple rule for making labour and investment choices,

shape the dynamics of macroeconomic variables like production, trade, industrial specialisation, wage disparities, and capital reallocation across industries. This approach contrasts with top-down equilibrium models, as it does not impose global optimality but instead lets economic structures evolve endogenously through heterogeneous agent decisions.

2.3 Production, wages, and returns

The aggregate labour and capital from citizens are used as the factors of production in each sector within the nation. The production or output of sector f in the country c is a function of the aggregate labour L and the aggregate capital K . These are matrices with dimensions that are given by the $n \times m$ where n and m are the number of countries and the number of sectors respectively. The aggregate choices of citizens regarding employment (L) and investment (K) in each sector define the total factor inputs available for production. These aggregate values are then used as inputs into sectoral production functions, which determine the total output of each country-sector combination.

Formally, let $L(c, f)$ and $K(c, f)$ denote the total labor and capital allocated to sector f in country c . These values are aggregated from individual citizen decisions:

$$L(c, f) = \sum_{i \in c} \mathbf{1}(l_i = f), \quad K(c, f) = \sum_{i \in c} \mathbf{1}(k_i = f) \quad (4)$$

where $\mathbf{1}(\cdot)$ is an indicator function that equals 1 if citizen i selects sector f and 0 otherwise. The total output ($Q(c, f)$) of sector f in country c is a function of aggregate labour (L) and capital (K). The model is flexible in accommodating different production function specifications, allowing for variations in returns to scale and input substitutability. In the simulations in the previous section, we use the Cobb-Douglas Production Function given below:

$$Q(c, f) = A(c, f)L(c, f)^\alpha K(c, f)^\beta \quad (5)$$

where $A(c, f)$ is the Total Factor Productivity (TFP), reflecting the technological efficiency of sector f in country c . α and β are output elasticities of labour and capital, respectively. This formulation ensures that sectoral productivity is endogenous, meaning that differences in technology, labor availability, and capital accumulation drive comparative advantages in production.

Given the production function, the wage rate ($W(c, f)$) and rate of return on investment ($R(c, f)$) are determined by the marginal productivity conditions:

$$W(c, f) = p(c, f) \frac{\partial Q(c, f)}{\partial L(c, f)}, \quad R(c, f) = p(c, f) \frac{\partial Q(c, f)}{\partial K(c, f)} \quad (6)$$

Note that $p(c, f)$ here refers to the price of the good f in country c . These prices are initialised exogenously but are changed dynamically after each simulation run. For purposes of numerical simulations and a more generalised formulation, we use a

finite difference approach to calculate wage where

$$W(c, f) = p \frac{\partial Q(c, f)}{\partial L(c, f)} \approx p \frac{Q(L + \Delta L, K) - Q(L, K)}{\Delta L} \quad (7)$$

Then we impose that returns on capital are paid out after the paying out the wage bill for production.

$$R(c, f) = \frac{pQ - WL}{K} \quad (8)$$

The wages and returns are used by agents to consume goods. In the simulations of the model, the demand for good f in country c , is calculated using a simple hyperbolic demand function in the form of:

$$D(c, f) = \frac{Y}{np(c, f)} \quad (9)$$

where the income is used in equal proportion to buy goods different sectors. The demand function can be formulated in multiple ways as well as has been a long and well studied sub-field in economics [].

In the absence of trade, the supply of any good $S(c, f)$ is given by the production of that good $Q(c, f)$. Given the demand and supply, we can create a mechanism to adaptively change the price $p(c, f)$.

There is a nominal good $f = 0$ and its price is fixed $p(c, 0) = 1 \forall c$. The nominal good is used as currency for trade and a measure of the value of other goods in the system.

$$p(c, f)(t + 1) = (1 + z\epsilon)p(c, f)(t) \quad (10)$$

where, at time $t = 0$, we have $p(c, f)(0) = 1$. ϵ is a constant that gives us the step size and z gives us the direction of change of the prices.

$$z = \begin{cases} +1 & \text{if } D(c, f) > S(c, f) \\ -1 & \text{if } D(c, f) < S(c, f) \end{cases} \quad (11)$$

2.4 Trade

We generalize the trade mechanism built in [20] which is considered a simple trade mechanism based on the price of goods in a 2-country 2-sector model. The generalized model offers a framework for trade which provides a robust trade mechanism for a system with n countries and m sectors. Trade occurs between two countries c and c^* when the following conditions hold:

1. Price of good p is cheaper in c : $p(c, f) < p(c^*, f)$
2. Stock of f in c is adequate: $S(c, f) > \delta$ where δ is the minimum amount of goods required to trade.
3. Buyer c^* has adequate liquidity in the form of nominal good $f = 0$: $S(c^*, 0) > \delta p(c, f)$

If these conditions are met, trade is executed:

$$T(c, c^*, f) = \begin{cases} \delta, & \text{if trade conditions hold} \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

Trade is a bilateral exchange so naturally $T(c^*, c, f) = -T(c, c^*, f)$.

$$S(c, f)(t) = Q(c, f)(t) + \sum_{c^*} T(c, c^*, f) \quad (13)$$

where S is the supply and T is a parameter that allows trade to be switched on or off. It is important to note that the second condition for trade could be removed by setting δ to be a proportion of total production rather than have it constant.

In a 2-country model, picking a trading partner (c, c^*) is to evaluate the conditions for trade above is trivial. However, in a n -country model, the order of trading of trades is important because goods produced including the nominal good may not be adequate to complete trades between each pair. In the model, we consider both randomised ordering of pairs and an adaptive mechanism in large n models which is outside the scope of this paper.

2.5 Capital Mobility

The capital mobility mechanism in the model allows for the dynamic reassessment and reallocation of capital investments by individual agents, leading to aggregate shifts in sectoral and national capital stocks. Each citizen initially allocates their capital to a randomly chosen sector within their country. However, with a probability p_{reassess} , individuals re-evaluate their investment decisions by comparing the rate of return on capital (R) across all sectors and, if capital mobility is enabled, across all countries. If capital mobility is restricted, citizens can only reallocate capital within their domestic economy, choosing the sector with the highest observed R . When mobility is enabled, individuals assess investment opportunities globally and can shift their capital to foreign markets if a sufficiently higher R is observed. This decision-making process follows a probabilistic rule, where capital is reallocated only if the new sector or country offers an R greater than the current investment by a factor of $(1 + d)$, where d represents a decision threshold set to 0 by default. The aggregate capital stock in each country-sector combination is thus formed endogenously based on individual agent behaviors, making capital accumulation highly dynamic and responsive to shifts in profitability. The redistribution of capital not only affects production but also alters wage rates and trade balances across economies.

2.6 Tariffs

In the model, tariffs are implemented using directly into the trading mechanism. Tariffs are exogenously input into the model as a 2-D array, T_r , of the form (exporting country, importing country, industry). When country c chooses its trade partner c^* , the first condition checked after a pairing has been made is to check if domestic prices are higher than the exporting country $p(c, f) > p(c^*, f)$. Here, the price of the exporter changes to $p(c^*, f) \rightarrow p(c^*, f)(1 + T_r(c^*, c, f))$ for c . The tariff imposes a deliberate increase in prices of the good from sector f imported from country c .

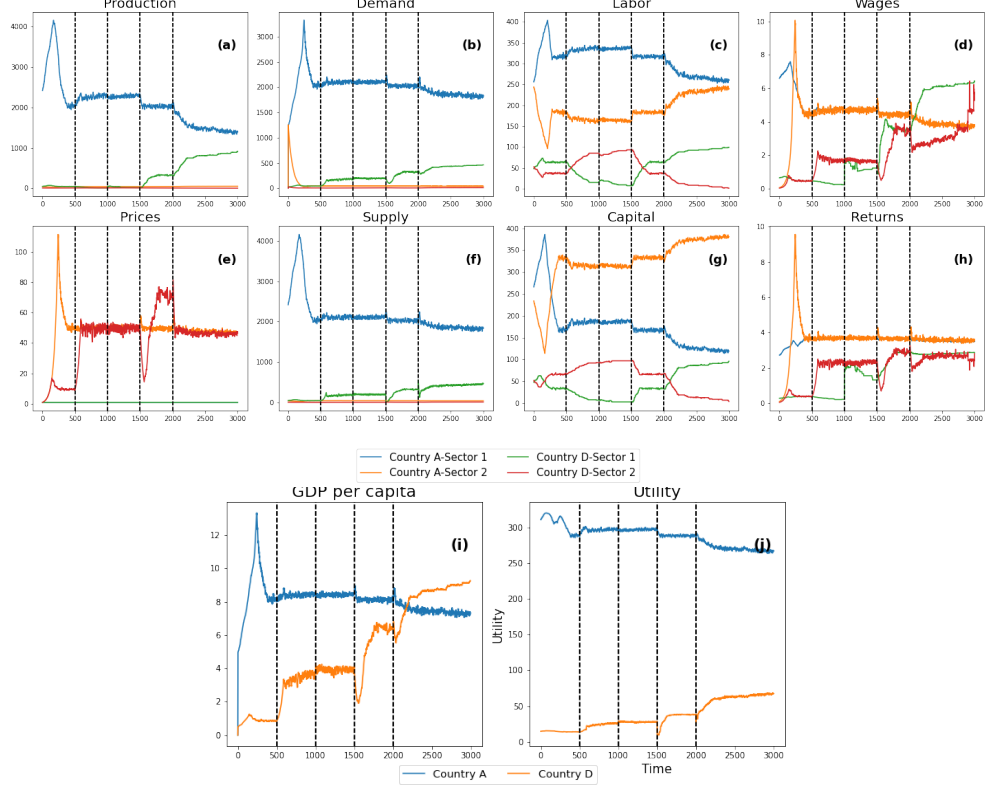


Fig. 2: Increasing returns to Scale: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. Population sizes of countries A and D are 500 and 100 respectively with initial productivity parameters $A^A = 1, 0.5$ and $A^D = 0.2, 0.2$. At $T = 500$ trade opens, at $T = 1000$ there is productivity shock introduced with $A_1^D = 1.5$. At $T = 1500$, trade is suspended and production continues in autarky and subsequently restarted at $T = 2000$. This is a simulation described in [20].

3 Theoretical Analysis

In this section, we simulate the model for stylized scenarios. We consider introducing increasing returns to scale and capital mobility in a 2 country 2 sector system. We also showcase the flexibility of the model by introducing tariffs and consider the impact of tariffs in a system with increasing returns to scale.

3.1 Increasing returns to scale

A natural extension of the H-O model was to suspend the assumption of constant returns to scale and introduce industries characterized by increasing returns to scale as done by [22]. This approach highlights the existence of multiple equilibria and how historical events and initial or "first mover" advantages can lock nations into stable but suboptimal patterns of production and trade. It contrasts with Samuelson's framework by showing how market forces alone cannot always be relied upon to produce optimal or equitable outcomes. We reproduce the simulation considered by [20] of this scenario with 2 nations — country A which is a developed nation with 500 citizens, and country D, a smaller, developing nation with 100 citizens. Each nation has two sectors: agriculture (named sector 1) and manufacturing (named sector 2) respectively. The agri-sector (or sector 1) follows decreasing returns to scale (typical of traditional sectors like farming), while the manufacturing sector (or sector 2) is characterized by increasing returns to scale (as seen in modern industries like electronics or automobiles). Initially, both countries differ in their productivity levels: the developed country is more productive in both industries with productivity coefficients $A_1^A = 1$ and $A_2^A = 0.5$, and thus a stronger comparative advantage in manufacturing (or sector 2), while the developing country has a slight comparative advantage in agriculture (or sector 1) with productivity coefficients $A_1^D = 0.2$ and $A_2^D = 0.2$. The scenario for the simulation is characterised by the following steps:

1. At $T = 0$: Citizens make labour and investment choices and nations produce and consume in autarky.
2. At $T = 500$: Nations trade and in the process increase, albeit marginally, the production in the sector with respective comparative advantage.
3. At $T = 1000$: Country D is given a productivity shock with A_1^D going from 0.2 to 1.5. The simulation shows that the productivity shock had little to no impact on the production or growth of the economy in country D even with the comparative advantage in sector 1.
4. At $T = 1500$: Trade is suspended and each country continues to produce in autarky. Country D produces goods with the increase in productivity and comparative advantage in sector 1.
5. At $T = 2000$: Trade is reopened and the impact of the productivity shock is realised and the country D grows its production in sector 1.

The increasing returns to scale is implemented here by having the output elasticity of labour parameter (α) in our Cobb-Douglas production function set to $\alpha_1 = 0.7$, $\alpha_2 = 0.4$ for both countries giving significant returns to every additional unit of labour in sector 1. This leads to the first-moved advantage for country A to overcome the comparative advantage held by country D after the productivity shock. The results of the simulation can be seen in figure 2. Among the many things here, it is vital to see the impact of the first-mover advantage in play. It shows how increasing returns to scale affect trade outcomes, emphasizing the critical role of policy in development strategies.

3.2 Capital Mobility

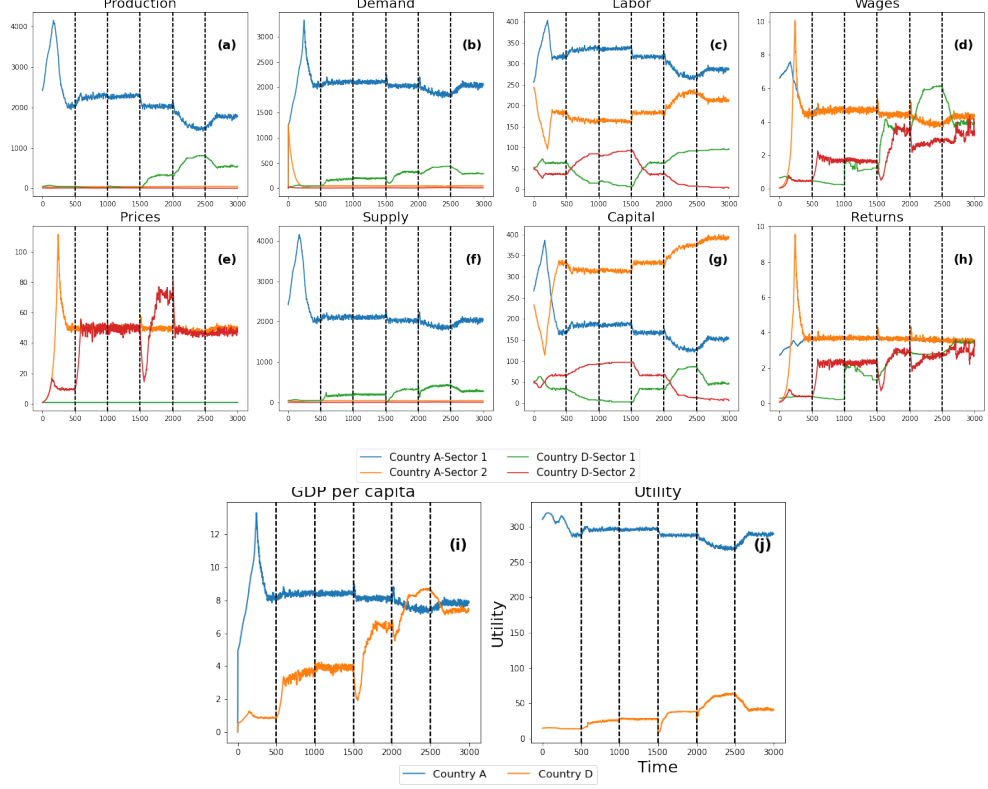


Fig. 3: Increasing returns to scale with Capital Mobility: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. Population sizes of countries A and D are 500 and 100 respectively with initial productivity parameters $A^A = 1, 0.5$ and $A^D = 0.2, 0.2$. At $T = 500$ trade opens, at $T = 1000$ there is productivity shock introduced with $A_1^D = 1.5$. At $T = 1500$, trade is suspended and production continues in autarky and subsequently restarted at $T = 2000$. At $T = 2500$, the simulation introduces capital mobility with citizens reassessing investment choices allowed to consider foreign investments.

A natural extension to the scenario above would be to relax the constraint of capital mobility that is explored by [23]. With a similar setup as [20], the authors introduce another event at $T = 2500$, introducing mobile capital, and thus allowing investment in sectors in foreign countries where returns are higher. In step 1 of the model introduced in the previous section, when reassessing investment options across sectors, citizens are allowed to invest their unit of capital in a sector in a foreign country if they offer a higher return. In the simulation in figure 3, we see that country D loses out on its growth trajectory after $T = 2500$ when capital mobility is introduced as capital flows out to sector 1 of country A. This result was shown in [20] and highlighted that in an industry with increasing returns, the country with the largest workforce in the sector

will attract a higher proportion of global capital. In this case, it is despite country D having a comparative advantage in the sector.

It is interesting to note here that the presence of increasing returns leads to multiple equilibria, making market outcomes path-dependent and often inefficient. This aligns closely with the idea originally presented in [24]. The development of industries with increasing returns—such as high-tech manufacturing—creates a similar effect where incumbents retain their advantage, making it hard for newcomers to compete without deliberate intervention. While in the simulation, the initial conditions of the TFP or $\mathbf{A}$ matrix shape long-term trade patterns, [24] argues that small historical events can lead to the dominance of one technology over another.

In the increasing returns regime, there exists the possibility of multiple equilibria. In our model, we can show this by running a simulation which shows that even under homogenous, equitable conditions, a system with countries following uniform growth patterns can develop into a system with a hierarchical structure with dominant economies emerging. The emergence of dominant economies in the model is primarily driven by increasing returns to scale. The stochastic processes in job and investment choices play a crucial role in reinforcing early advantages where small initial differences (due to randomness) when trade opens get amplified over time due to self-reinforcing advantages. We explore the scenario with $n=5$ countries and $m=3$ sectors in figure 4 where we consider equitable homogenous conditions with each country having the same TFP values, α , β values. We select sector m^* having increasing returns to scale such that $\alpha_{m^*} = \beta_{m^*} = 0.7$ while $\alpha_{m \neq m^*} = \beta_{m \neq m^*} = 0.4$. In autarky, we see all countries growing uniformly. When trade opens at $T = 500$, one country that happens to accumulate a slight edge through random fluctuations in L and K in sector 1 accumulates a disproportionate share of the high-productivity sector. This result aligns well with the theoretical insights from [24] on increasing returns and lock-in where small fluctuations can accrue an advantage which gets amplified due to the positive feedback in the system.

3.3 Tariffs

We introduce a simple tariff system using a tariff matrix that adjusts the relative price of imported goods to explore how tariffs can impact the trade flows. We show the integration of tariffs in the model above to explore. Tariffs influence the trade in particular sector between any pair of countries. The importing country increases the price of the good imported, thereby discouraging the trade. Tariffs in this context may create temporary distortions, increasing the cost of trade between specific countries and reducing trade volume, but they do not fundamentally alter the underlying equilibrium-driven dynamics. This can be seen, for example, in the Samuelson case in figure A3. We introduce a scenario with country A imposing a tariff of 100% on country B's manufacturing sector (sector 2) at time $T = 1500$. As a result, there is an increase in domestic production, wages, and returns in sector 1 for country A until the system stabilises to a relatively less disparate state of production. With the prices of $p_2^B \rightarrow p_2^B(1 + t_r)$ where $t_r = 1$ is the tariff rate. With the increase in prices, the demand from domestic produce of sector 2 in country A increases until it reaches the

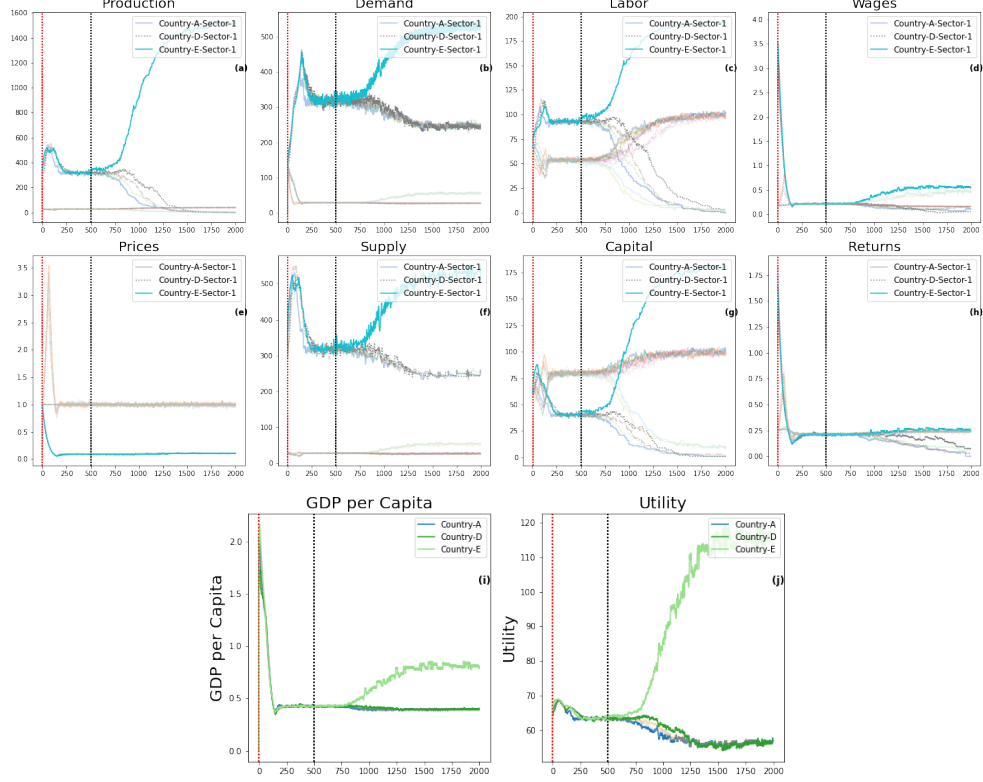


Fig. 4: Increasing returns in homogeneous conditions: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. Here we have $n = 5$ countries and $m = 3$ sectors with homogeneous conditions for production. Each country has a population of 200 with uniform initial productivity parameters $A_m^n = 1$ for all countries and sectors. For one sector, $m = 1$, $\alpha_1^n = \beta_1^n = 0.7$ while for all other sectors $\alpha_{\neq 1}^n = \beta_{\neq 1}^n = 0.4$ which sets that sector 1 has increasing returns to scale in all countries where all other sectors have decreasing returns to scale. At $T = 500$ trade opens, the simulation grows significantly having gained advantages in sector 1 while others struggle to grow.

new import price. However, under increasing returns to scale, a small temporary loss in trade may lead due to a tariff imposed that may have a significant impact on global trade. With increasing returns, early movers lock in advantages that make it difficult for new entrants to compete. Tariffs provide a protected window for industries outside the dominant country to grow, accumulate knowledge, and reduce production costs over time. This can lead to a permanent shift in global trade dynamics, where a country that was initially uncompetitive in a sector can, through temporary protection, develop industries that would not have emerged under free trade.

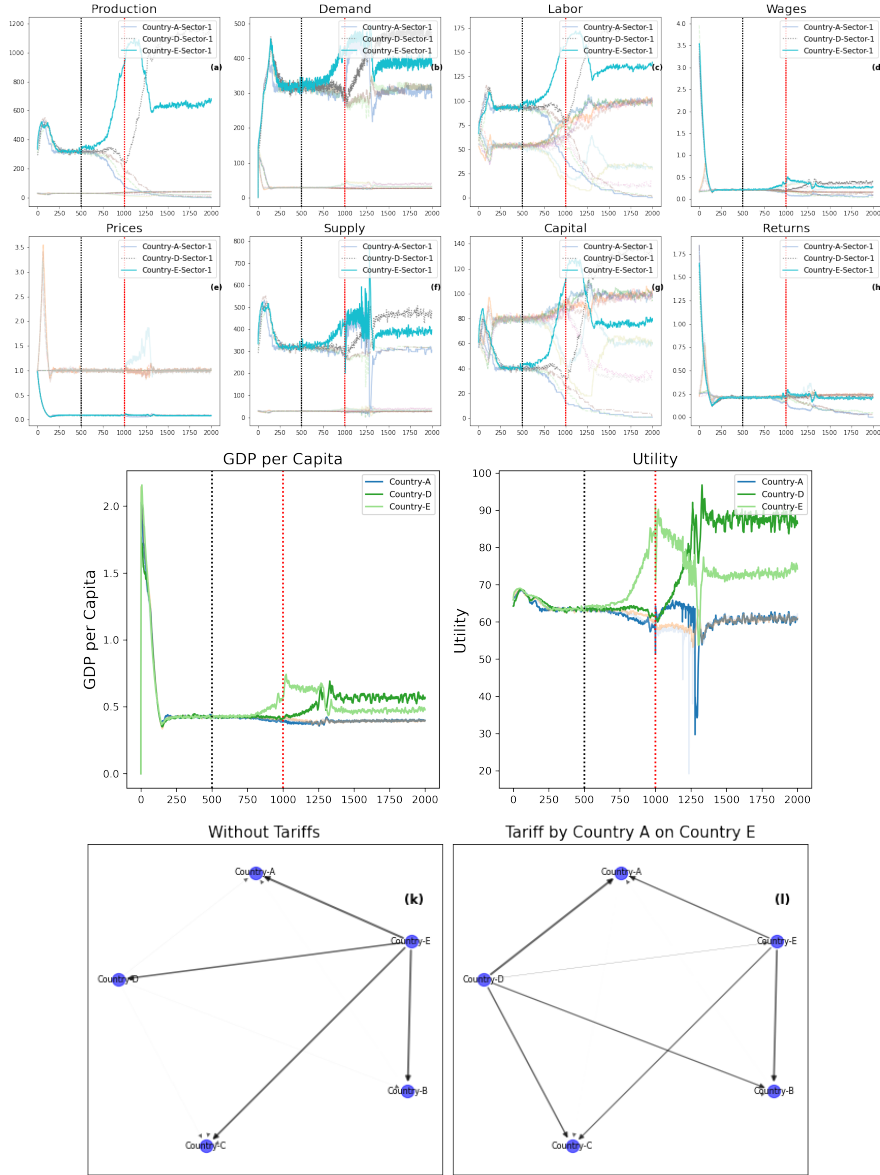


Fig. 5: Increasing returns with Tariffs introduced: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. This is the same setup as simulation described in figure 4. At $T = 1000$, we introduce a 50% tariff on the imports of goods in sector 1 from country E to country A. Only values in sector 1 for countries A, D, E are highlighted in the figure. Figures (k) and (l) show the trade networks before and after the imposition of tariffs, respectively. The width of the edges highlight the strength of total trade between the pair of countries across all the sectors.

We explore this scenario in figure 5. In order to show the effect of tariffs in the case of increasing returns, we use the same setup as described in 4. In this particular simulation, we see that country E dominates the production and export of goods from sector 1 once trade opens at $T = 500$. We introduce a tariff at $T = 1000$ here where in this case the country A (chosen arbitrarily) imposes a tariff of 50% on the exports of sector 1 in country E, which dominates its production. The imposition of the tariff increases the price of goods from sector in country E to be exported to country A. Hence, country A chooses an alternative source for its demand from goods in sector 1. Given that there is immediate demand, country A cannot increase its production immediately and thus looks around to pick another trading partner. This is a random choice, and the chosen trading partner benefits from the imposition of the tariff. The trading partner increases its production to fulfill the export demands and alters the equilibrium in which the system settles. We look at the total trade flow between countries with and without the tariffs in figure 5 (k)(l) respectively. Before the imposition of the tariffs, it is evident in figure 5(k) that global exports are dominated by country E. Country D also exports goods to A, B, and C but the size of the exports is much smaller when compared to that of E. After the imposition of tariffs, country D becomes the dominant exporter of goods in sector 1 and the trade network becomes divided as seen in figure 5 (l).

4 Empirical Calibration and Analysis

We have demonstrated that agent-based modeling can create numerical simulations of theoretical outcomes and allow deeper investigation by relaxing certain assumptions (such as capital mobility or increasing returns to scale). The results seen so far are designed to reproduce analytical results. We provide an extremely generalised agent-based model that can produce results while offering a modular framework to run experiments on a generalised $n \times m$ case while incorporating different functions to account for the behaviour of both citizens and nations. The parameters and functions used for the simulations shown above are presented in the table 1. To further assess the model's applicability to real-world scenarios, we develop a method for calibrating it using empirical trade data. This calibration ensures that the model not only reproduces theoretical predictions but also aligns with observed global trade patterns.

To achieve this, we employ Multi-Region Input-Output (MRIO) tables from the EORA database ¹ [25]. The MRIO tables provide detailed sectoral trade flow data across 187 countries and 26 industries, comprising: (i) a transactions table (T), which captures inter-sectoral trade between country-sector pairs, (ii) a final demand table (Y), which tracks direct consumption by households, governments, and export markets, and (iii) a value-added table (V), which accounts for labor, capital, taxes, and other sources of income across industries. For simplicity, we aggregate the 26 industrial sectors into three broad categories: Primary, Secondary, and Tertiary, as detailed in Table A1 in the appendix. The calibration focuses specifically on the final demand

¹See <https://worldmrrio.com/eora26/> for more details

Parameter	Description	Default Value
n_c	Population (static)	Initialized
n	Number of countries in the simulation	Initialized
m	Number of products in the simulation	Initialized
$\mathbf{A}$	Technology matrix for each nation-industry pair	Initialized
α	Labor elasticity parameter in production	Initialized
β	Capital elasticity parameter in production	Initialized
$p_reassess$	Probability of a worker reassessing job choice per step	0.004
production_function	Production function	Cobb-Douglas Production
wages_return_function	Method for computing wages and returns	Marginal production
utility_function	Method for computing utility	Geometric Mean of Supply
pricing_function	Method used for price adjustments	Demand-Supply Gap
trade_function	Method for finding trade partners	Randomized
δ	Rate of trade volume changes per step	0.01
ϵ	Rate of price changes per step	0.02

Table 1: List of parameters: The table shows the different parameters and specific functions used in the model. The ones highlighted in bold are matrices initialised in the simulations in figures [A1](#), [A2](#), [2](#), [3](#).

table (Y), representing trade flows that directly result from consumption and production decisions, aligning closely with the behavioral assumptions of our agent-based model.

The calibration process focuses on three key economies— USA, Mexico, and China—chosen due to their significant and interdependent trade relationships. The USA represents a large, developed economy with strong global trade linkages, Mexico serves as an emerging market with close economic ties to the USA through regional trade agreements (e.g., USMCA), and China represents a major global manufacturing hub with extensive trade flows across multiple sectors. This selection captures a diverse range of economic structures, allowing the model to be tested under varying production and consumption patterns. Broadly, the calibration processes follows the steps below:

1. We select the countries, and aggregate trade flows across the three broad industrial sectors from the MRIO tables. The aggregation from the final demand table and the trade flow table output from the model are described in the schematic in figure [6 A](#).
2. The populations of nations are normalised for computational purposes. To ensure a proportional scaling of country populations, we normalize them such that the smallest country (Mexico) has a population of 100.
3. The model is initialised with country specific values for total factor productivity (TFP, $\mathbf{A}$ matrix), elasticity of labor (α), and elasticity of capital (β). The elements of the α matrix are drawn randomly from a uniform distribution within the range $[0, 1]$, while β is set as $1 - \alpha$ to enforce constant returns to scale.

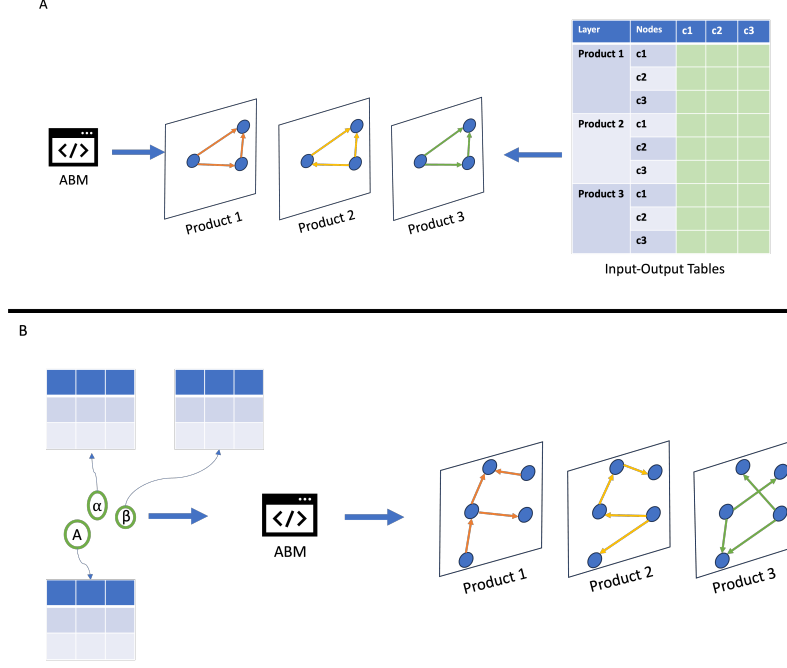


Fig. 6: Schematic descriptions of the calibration process: Plot A depicts shows that the trade flow data obtained from the ABM, and the aggregated MRIO tables are as a multi-layer network with sector representing a layer. We also aggregate MRIO tables in a similar way.

4. The model is run with constant returns to scale, open trade, and capital mobility and run for $T = 2000$ iterations. The resulting trade flow matrix from the simulation is normalized², and its accuracy is assessed by computing the root mean square error (RMSE) relative to the normalized trade flow matrix.
5. Based on the error, new values of A , α are selected using meta-heuristic method and the trade flow matrices are evaluated again.

To calibrate our agent-based model to real-world trade flow data, we employed NGOpt, an advanced black-box optimization framework introduced in Nevergrad, a derivative-free optimization platform³[26]. NGOpt (Nevergrad Optimization) is a meta-heuristic approach designed for high-dimensional, complex optimization problems where gradient-based methods are either infeasible or inefficient. It integrates a diverse set of optimization techniques, including evolutionary algorithms, Bayesian optimization, differential evolution, and other evolutionary optimisation algorithms,

²We use a simple min-max scaling for normalising the entries of the matrix into a range of $[0, 1]$ given by $\hat{X} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$

³See <https://facebookresearch.github.io/nevergrad/> for more details

allowing it to adapt dynamically to different types of objective functions and constraints. The approach continues to search in the parameter space for values of A and α that produce trade flow matrices as shown in schematic in figure 6B.

Year	Correlation
2014	0.85
2015	0.83
2016	0.81
2017	0.80
2018	0.82
2019	0.81
2020	0.83

Table 2: The correlation between the normalised simulated and empirical trade networks for all years between 2014-2020.

Using Multi-Region Input-Output (MRIO) data from 2019, the agent-based model was run for 2,500 different sets of parameters selected using the meta-heuristic optimisation framework, achieving a root mean square error (RMSE) < 0.006 and an overall correlation of $> \approx 0.8$ between the simulated and empirical trade networks. We assume sectorally uniform productivity levels within each country to focus on cross-country differences. The USA exhibits the highest TFP across all sectors at approximately 0.815, suggesting a relatively balanced and efficient production structure in Primary, Secondary, and Tertiary industries. In contrast, Mexico’s TFP is the lowest at 0.122, indicating lower overall productivity despite its role in global manufacturing chains. China sits between the two, with a moderate TFP of around 0.315 across all sectors, reflecting its transitional status between developing and industrialised economies. The uniformity of TFP values across sectors simplifies the analysis, allowing clearer interpretation of how trade outcomes are shaped by macro-level productivity differences rather than sector-specific specialisation. We calibrate values for the elasticities of labour (α) and capital (β) for each of the country-sector combinations. These values indicate that the USA’s primary sector is highly labour-intensive ($\alpha \approx 0.86$), while its Secondary sector relies more on capital. Mexico shows moderate labour intensity across sectors, with slightly higher emphasis on the primary sector. China’s Secondary sector stands out with a high labour elasticity ($\alpha \approx 0.93$), suggesting a strong reliance on labour in manufacturing. The capital elasticity (β), calculated as $(1 - \alpha)$, complements these findings, highlighting the varying degrees of capital dependence across countries and sectors. These calibrated parameters provide insights into the structural differences in production processes among the three countries, reflecting their unique economic compositions and development stages.

In terms of trade flows, the largest trade flow predicted is between USA to China across all the sectors with the US exporting secondary and tertiary sector goods while China exporting primary sector goods. While these results indicate a strong alignment

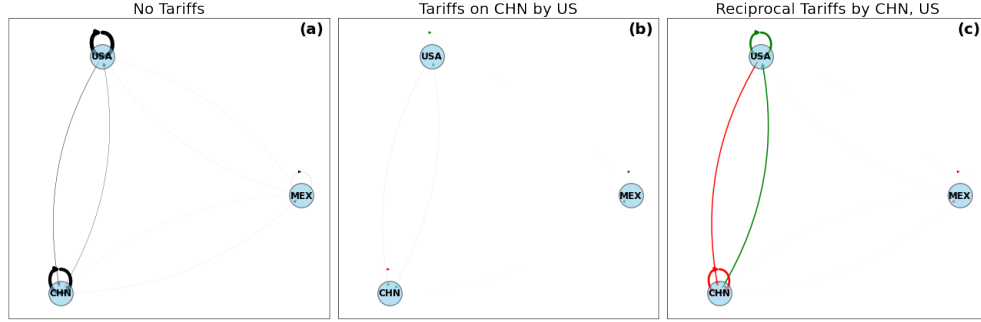


Fig. 7: Total trade flow network with tariffs: Plot (a) shows the trade flows without any tariffs. Plot (b) shows the change in trade flows under the scenario where USA imposes tariffs of 100% on Chinese exports. As you can see, the changes are relatively small. Any positive changes in trade flows are represented by lines in green, and any negative changes in trade flows are represented by lines in red. Plot (c) shows the change in trade flows under the scenario where China has a reciprocal tariff on US exports.

between the model and observed trade patterns, the retrieved parameter values should be interpreted with caution, as they do not necessarily reflect real-world economic fundamentals. Several unmodeled factors—such as sector-specific increasing returns to scale, taxation policies, labour market efficiency, and regulatory barriers—can influence trade flows and impact the estimated parameters. The primary objective of this calibration is not to derive precise parameter estimates but to assess the model’s ability to replicate observed trade patterns using a simplified agent-based framework.

4.1 Tariffs and reciprocal tariffs

With the calibrated model, we explore a potential scenario of how tariffs can affect the trade flow in this simple 3-country, 3-sector scenario. We simulate the trade model with trade and capital mobility enabled and introduce a tariff of 100% on Chinese exports into the US. Interestingly, the results, depicted in figure (7), reveal a slight overall increase in trade among the three countries compared to the no-tariff baseline. This counterintuitive outcome suggests that while the tariff directly targets Chinese exports to the U.S., it may inadvertently stimulate trade flows among the countries, potentially due to shifts in comparative advantages or trade diversion effects. Subsequently, we introduce a reciprocal 100% tariff by China on U.S. exports. This adjustment leads to a notable asymmetry: U.S. exports to China decrease significantly, while Chinese exports to the U.S. remain relatively stable. Even in this simplified model of international trade, it is interesting that we find this counterintuitive result: the imposition of tariffs does not uniformly suppress trade but instead reshapes trade patterns based on underlying economic structures and relationships. We saw this in the scenario introduced in the earlier section 3.3 represented in figure (5). These findings underscore the complex and often counterintuitive nature of international trade dynamics, where

the imposition of tariffs does not uniformly suppress trade but instead reshapes trade patterns based on underlying economic structures and relationships.

5 Discussion

The model by [20] was highly stylised, and had a limited scope, with the objective of setting up a simple model to simulate comparative advantage in an agent-based setting leveraging the well-studied H-O model. In this paper, we expand on this approach by making two key contributions: (i) we build generalised model which can accommodate a larger system of agents, countries and sectors, while also showing the flexibility of the approach in incorporating mechanisms such as tariffs as evidenced in figures 4, 5 (ii) we show that the model can also be calibrated to real-world trade flows data. The model can be used as a policy laboratory, allowing for the simulation of theoretical scenarios studied, as well as a tool used to study real-world trade networks under different policy scenarios.

The first set of results simply showcase the model capacity to reproduce important theoretical results such as Samuelson’s Outsourcing analysis [21], increasing returns and retainable industries by Gomory and Baumol [22] and Herman Daly’s capital mobility [23] as shown by [20]. We show that technological improvements in one country can dramatically reshape global trade patterns. By incorporating a third country in Samuelson’s set-up (see figure A2), we show that a productivity shock in one industry in a specific country does not remain isolated—it alters trade balances, wages, and capital flows across multiple economies.

One of the most striking findings of the model is that in a homogeneous system (where all countries start with identical conditions), increasing returns to scale leads to extreme trade imbalances. Over time, one country tends to dominate the system, accumulating a disproportionate share of industrial production and capital inflows. This aligns with the work of [24] on lock-in and path dependence, where small initial advantages lead to long-term dominance. This finding suggests that policy interventions (such as strategic tariffs or subsidies) may be required to maintain competitive trade dynamics. We introduce tariffs into the system that can exogenously affect the price at which two countries trade. Under decreasing or constant returns to scale, tariffs do not necessarily provide a qualitative change to the equilibrium dynamics of the model. However, under increasing returns to scale, can create path-dependent shifts that can permanently alter trade networks dynamics.

The calibration of the model to real-world trade data is an important step toward making ABMs as a complementary tool to Computable General Equilibrium (CGE) models. To be useful, trade policy models must capture highly complex phenomena: changes in trade relationships have multiple effects through different channels. Among other things, CGE models produce a baseline for the economy, followed by a shock, and consequently, the final steady-state outcome. It does not, however, present results on how the economy transitions through time to the final steady-state. ABMs, on the other hand, can provide a numerical simulation based tool that offer more flexibility incorporating stochastic, adaptive processes and allow for the study of systems not necessarily in equilibrium. However, they are not always suitable for large-scale empirical calibration. We show that the model can reasonably replicate observed trade patterns, paving the way for further development of ABMs as practical tools for trade policy analysis.

The model, while providing valuable insights into global trade dynamics, operates with several key limitations, each of which presents an opportunity for future research. First, while the model builds on the Heckscher-Ohlin (H-O) framework, modern trade theories emphasize heterogeneous firms within nations rather than treating entire sectors as homogeneous entities. In this model, the behavior of nations is simulated at the sectoral level, assuming all firms within a sector exhibit identical characteristics. Although the model is designed with the capacity for adaptive processes, the simulations presented in this paper assume static economic structures.

For instance, demographics remain fixed—there is no population growth, labor mobility, or skill acquisition over time. Citizens make labour and investment decisions based on perfect information about wages in all sectors, which eliminates information frictions and the possibility of unemployment. We assume fixed sectoral productivities and elasticities, which removes the potential for technological innovation. Trade relationships are formed randomly, with no historical trade dependencies, geographical proximity effects, or trade agreements influencing partner selection.

Addressing these limitations in future iterations of the model could significantly enhance its realism and applicability. Introducing heterogeneous firms, evolving labor markets, imperfect information, and endogenous technological change would allow the model to better capture real-world trade patterns. Furthermore, incorporating geopolitical and spatial trade factors—such as transport costs, regional trade agreements, and historical trade dependencies—would provide a richer understanding of global trade network evolution.

6 Conclusion

The contributions of this paper are as follows: (i) First, we build a generalised, flexible model introduced by [20], allowing n countries, m sectors. We showcase the flexibility of the model to incorporate new elements such as bi-lateral tariffs on trade. (ii) Second, we showcase the capacity of the model to reproduce the theoretical results highlighted above. In addition, we show the behaviour of the model for homogenous agents with equitable conditions under increasing returns to scale. (iii) Finally, we show that the model can also be calibrated to real-world trade flows data and thus can be an interesting tool for studying the impacts of trade policy. Specifically, we fit the model to the data subsetting from the EORA database [25], using data for three major economies (USA, Mexico, and China) across three broad economic sectors (primary, secondary, and tertiary). This validation exercise highlights the potential of agent-based trade models not only for exploring theoretical questions but also for policy-oriented applications. Unlike existing agent-based trade models, our framework is fully modular, allowing for customization of functional forms for production, wage determination, investment behavior, and trade mechanisms. The model simulates economic dynamics from the bottom up, beginning with employment and investment decisions at the individual level and culminating in aggregate trade flows between nations.

Supplementary information. The data used for the calibration of the model can be found in the EORA26 database where data for the years 2014-2017 are available

for free. See <https://worldmrio.com/eora26/> for more details. The code used for the project can be found in the repository: https://github.com/Nk173/Trade_ABM.

References

- [1] Heckscher, E.F., Ohlin, B.: Heckscher-Ohlin Trade Theory. MIT Press, Cambridge, Mass. (1991)
- [2] Krugman, P.R., Obstfeld, M., Melitz, M.J.: International Economics: Theory & Policy. Pearson Addison-Wesley, Boston, MA (2022)
- [3] Grossman, G.M., Helpman, E.: Technology and trade. Handbook of international economics **3**, 1279–1337 (1995)
- [4] Krugman, P.R.: Increasing returns, monopolistic competition, and international trade. J. Int. Econ. **9**(4), 469–479 (1979)
- [5] Costinot, A.: On the origins of comparative advantage. J. Int. Econ. **77**(2), 255–264 (2009)
- [6] Leontief, W.: Domestic production and foreign trade; the american capital position re-examined. Proc. Am. Philos. Soc. **97**(4), 332–349 (1953)
- [7] Feenstra, R.C., Lipsey, R.E., Deng, H., Ma, A.C., Mo, H.: World trade flows: 1962-2000 (11040) (2005)
- [8] Krugman, P., Others: Scale economies, product differentiation, and the pattern of trade. Am. Econ. Rev. **70**(5), 950–959 (1980)
- [9] Krugman, P.R.: Intraindustry specialization and the gains from trade. J. Polit. Econ. **89**(5), 959–973 (1981)
- [10] Melitz, M.J.: The impact of trade on intra-industry reallocations and aggregate industry productivity. Econometrica **71**(6), 1695–1725 (2003)
- [11] Epstein, J.M.: Generative Social Science. Princeton University Press, ??? (2012)
- [12] Axtell, R.L., Farmer, J.D.: Agent-based modeling in economics and finance: Past, present, and future. J. Econ. Lit., 1–101 (2022)
- [13] Chen, S.-H., Chang, C.-L., Du, Y.-R.: Agent-based economic models and econometrics. Knowl. Eng. Rev. **27**, 187–219 (2012)
- [14] Dosi, G.: The Foundations of Complex Evolving Economies: Part One: Innovation, Organization, and Industrial Dynamics. Oxford University Press, Oxford, England (2023)
- [15] Anderson, P.W., Arrow, K.: The Economy as an Evolving Complex System. CRC

Press, Boca Raton, FL (1988)

- [16] Tesfatsion, L.: Economic agents and markets as emergent phenomena. *Proc. Natl. Acad. Sci. U. S. A.* **99 Suppl 3**(Suppl 3), 7191–7192 (2002)
- [17] Catullo, E.: An agent based model of monopolistic competition in international trade with emerging firm heterogeneity. *J. Artif. Soc. Soc. Simul.* **16** (2013)
- [18] Alves Furtado, B., Eberhardt, I.D.R.: A simple agent-based spatial model of the economy: Tools for policy. *J. Artif. Soc. Soc. Simul.* **19**(4) (2016)
- [19] Chliaoutakis, A., Chalkiadakis, G.: An agent-based model for simulating inter-settlement trade in past societies. *J. Artif. Soc. Soc. Simul.* **23**(3) (2020)
- [20] Gulden, T.R.: Agent-based modeling as a tool for trade and development theory. *J. Artif. Soc. Soc. Simul.* **16**(2) (2013)
- [21] Samuelson, P.A.: Where ricardo and mill rebut and confirm arguments of mainstream economists supporting globalization. *J. Econ. Perspect.* **18**(3), 135–146 (2004)
- [22] Gomory, R.E., Baumol, W.J.: *Global Trade and Conflicting National Interests*. MIT Press, Cambridge, Mass. (2022)
- [23] Daly, H.E., Farley, J.: *Ecological Economics, Second Edition: Principles and Applications*. Island Press, Washington, D.C., DC (2011)
- [24] Arthur, W.B.: Competing technologies, increasing returns, and lock-in by historical events. *Econ. J. (London)* **99**(394), 116 (1989)
- [25] Lenzen, M., Kanemoto, K., Moran, D., Geschke, A.: Mapping the structure of the world economy. *Environ. Sci. Technol.* **46**(15), 8374–8381 (2012)
- [26] Trajanov, R., Nikolikj, A., Cenikj, G., Teytaud, F., Videau, M., Teytaud, O., Eftimov, T., López-Ibáñez, M., Doerr, C.: Improving nevergrad’s algorithm selection wizard NGOpt through automated algorithm configuration. In: *Lecture Notes in Computer Science. Lecture notes in computer science*, pp. 18–31. Springer, Cham (2022)

Appendix A Theoretical results and extensions

We present the results of our generalized agent-based trade model, highlighting its ability to replicate theoretical insights such as Samuelson’s Outsourcing scenario [21] while extending the analysis by simulating the scenario for a third country.

A.1 Samuelson Analysis of Outsourcing

We begin with the simulation of Samuelson’s analysis of outsourcing introduced in [21]. This features two nations, country A and country B, that represent the economies of USA and China, and two production sectors 1 and 2 that represent agriculture and manufacturing industries (named wine and cloth in the original paper). As described in figure 1, we run the simulation using standard formulations for calculations at each step involved in the simulation as detailed in the section 2, namely, employment, investment, production, wages, return on capital, trade, pricing, demand (or consumption preferences), and utility. The results of the simulation are shown in figure A1.

The step-up is as follows: Country A has 100 citizens with sectoral productivities of 0.5 and 2 while Country B has 1000 citizens but with lower productivities of 0.2 and 0.05 respectively. Despite the disparity in per capita productivity, total production is roughly equal due to the difference in population sizes. Production is modeled using Cobb-Douglas production functions for each sector with the assumption of constant returns to scale. Each citizen has a unit of labour and capital to deploy in the economy and the simulation begins by randomly assigning each citizen to a sector for employment and to a sector for their investment. We simulate $T = 2000$ time steps and beginning in a state of autarky where each nation produces and consumes without trading. After $T = 500$ rounds, trade opens, allowing the nations to specialize based on comparative advantage where country A specializes in the manufacturing sector, while country B focuses on the agriculture sector, leading to significant utility gains for both. The model shows that under free trade, the countries enjoy similar levels of prosperity. A shock is introduced after $T = 1000$ time steps in the form of rapid technological innovation in country B’s manufacturing sector (i.e. the productivity goes from 0.05 to 0.8), a sector where the Country A had a comparative advantage previously. Under the current conditions, trade would make Country B better off and the Country A worse off. When Country B experiences a productivity boost in its cloth industry, its utility doubles, while Country A utility reverts to autarky levels. Here, utility is the geometric mean of the total supply in the country. This technological change eliminates the incentive for trade since both countries can meet their needs more efficiently on their own. This scenario highlights how outsourcing and technological improvements in one country can harm another by reducing trade benefits. Here wages and returns to capital are calculated based on marginal productivity. When trade opens, wages and returns rise in the specialized industries, which naturally direct labour and capital toward these sectors. The demand is calculated using a hyperbolic demand function each agent spends an equal proportion of their income on each good. The demand (and supply) for goods converge as trade opens up and prices equalise in

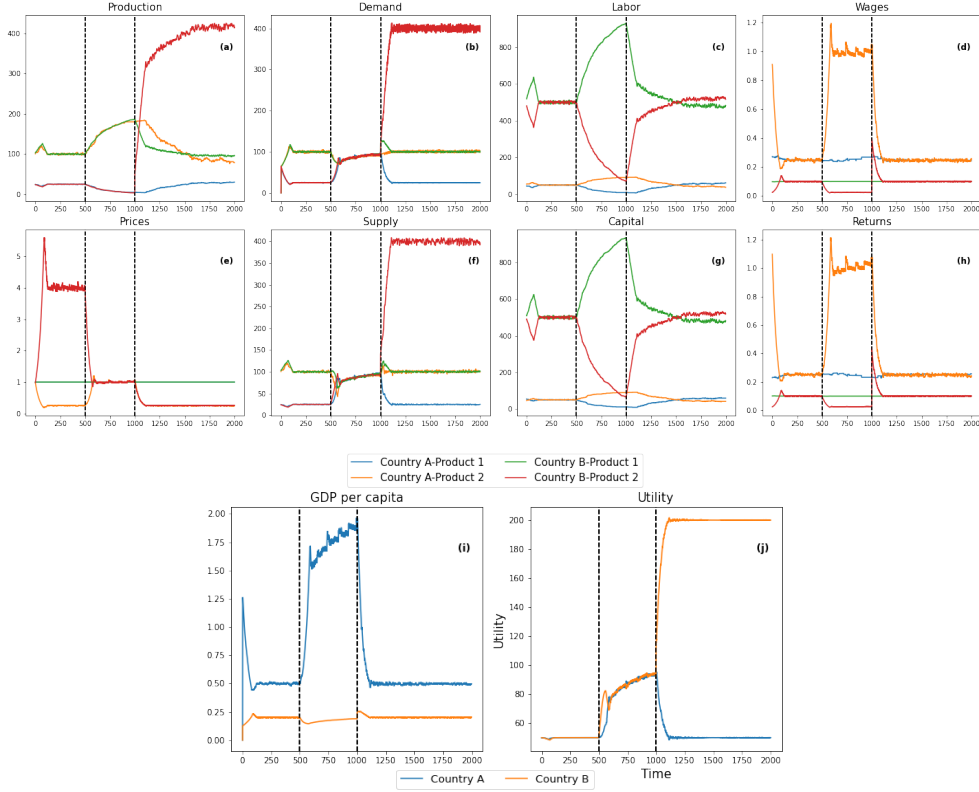


Fig. A1: Samuelson analysis of Outsourcing with 2 countries and 2 sectors: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. Population sizes of country A and country B are 100 and 1000 respectively with country A having productivity factors of $A_1^A = 0.5$, $A_2^A = 2$, $A_1^B = 0.2$, $A_2^B = 0.05$. Country A has a competitive advantage in sector 2 and country B in sector 1. At $T = 500$, trade between the two countries open and at $T = 1000$, a productivity shock is introduced with A_2^B goes from 0.05 to 0.8.

the economy. Section 2 provide more details on the analytical setup of the model. The simulation shown in figure A1 (a-h) shows the simultaneous dynamics of the various economic indicators for all combinations of country-sectors while the figure A1 (i),(j) give the aggregate national indicators.

There are several discussions, both theoretical as well empirical, that followed Samuelson's analysis of outsourcing. This argument was a departure from traditional Ricardian trade models, which generally assume that all nations benefit from trade due to comparative advantage. The primary result here is the ability of the model to showcase the result using a numerical simulation that showcases the simultaneous evolution of different economic indicators for this simple system. To showcase the ability

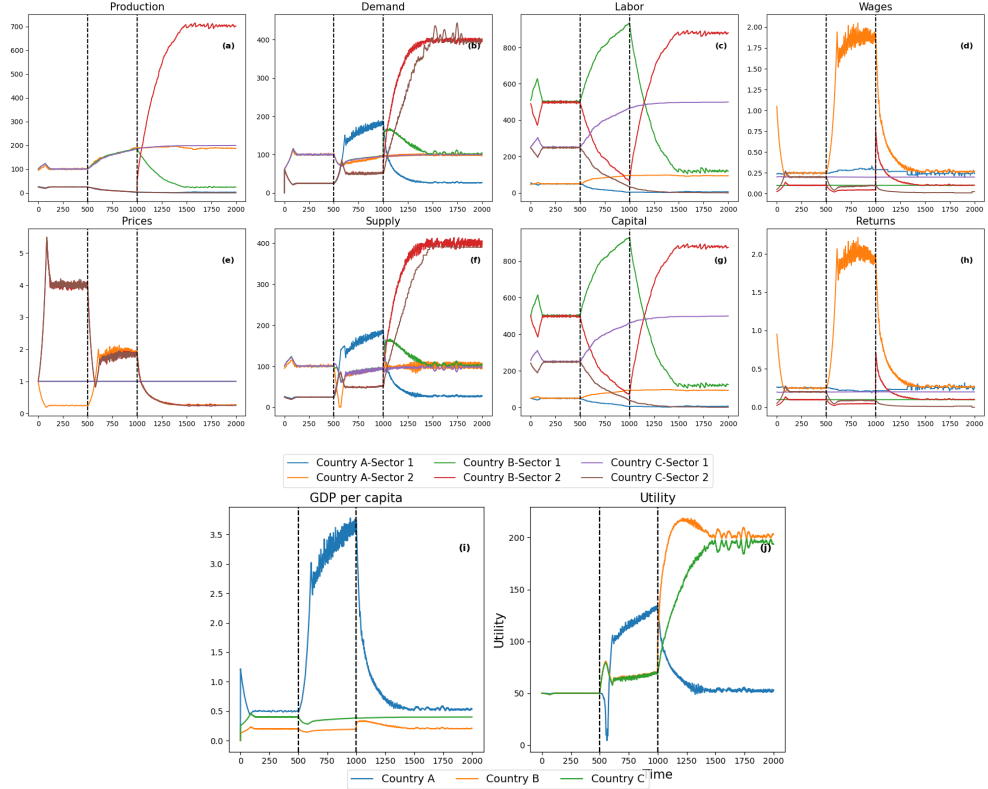


Fig. A2: Samuelson analysis of Outsourcing with 3 countries and 2 sectors: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. Population sizes of country A, B, C are 100, 1000, 500 respectively with country A having productivity factors of $A_1^A = 0.5$, $A_2^A = 2$, $A_1^B = 0.2$, $A_2^B = 0.05$ and $A_1^C = 0.2$, $A_2^C = 0.05$. Country A has a competitive advantage in sector 2 while countries B, C have similar productivities sector 1 and are in competition under free trade. At $T = 500$, trade between the countries open and at $T = 1000$, a productivity shock is introduced with A_2^B goes from 0.05 to 0.8.

of the model to accomodate higher levels of complexity, we introduce a third country, Country C with the same relative productivity parameters as Country B (let us assume the third country is similar to India). There are now 3 countries and 2 sectors in the system which introduces an element of competition under free trade. Countries B and C here have similar competencies whereas country A has a comparative advantage in a sector.

The results in figure A1 show that under free trade all countries benefit but the country A benefits significantly more than the other two. With the introduction of the same technological shock as before, the countries that were in competition earlier, now

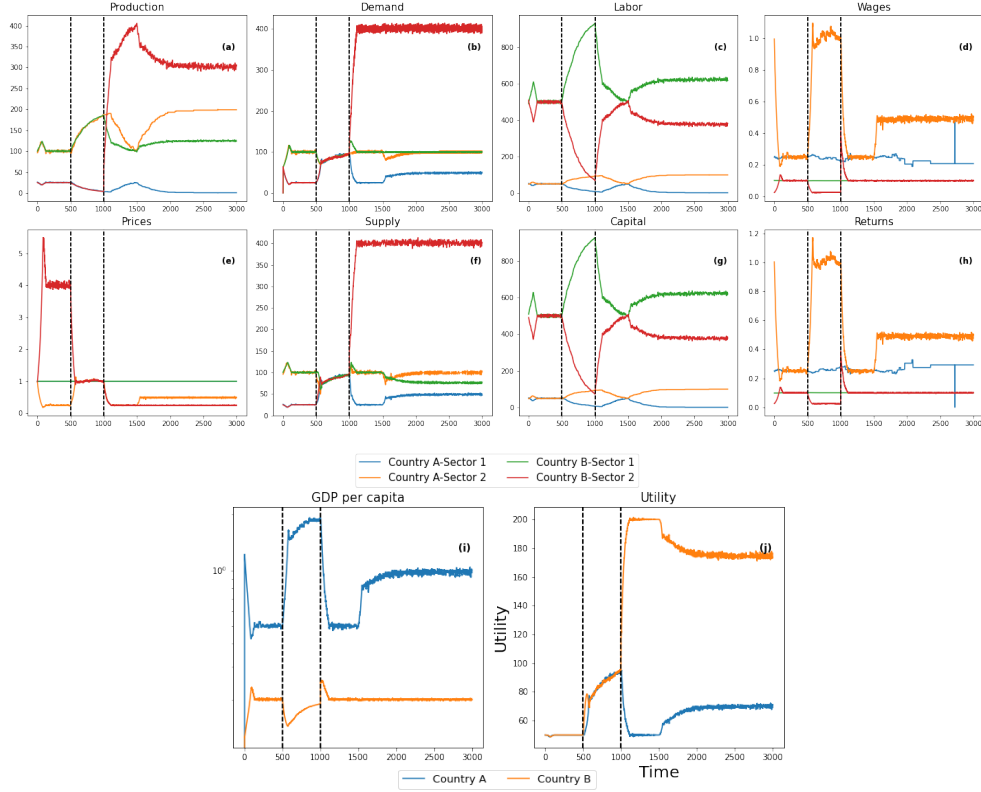


Fig. A3: Samuelson Outsourcing scenario with tariffs: Here plots (a)-(j) show the model output for (a) production, (b) demand, (c) labour, (d) wages, (e) Prices, (f) supply, (g) capital, (h) returns, (i) income per capita, and (j) utility respectively. Population sizes of country A and country B are 100 and 1000 respectively with country A having productivity factors of $A_1^A = 0.5$, $A_2^A = 2$, $A_1^B = 0.2$, $A_2^B = 0.05$. Country A has a competitive advantage in sector 2 and country B in sector 1. At $T = 500$, trade between the two countries open and at $T = 1000$, a productivity shock is introduced with A_2^B goes from 0.05 to 0.8. Here, we introduce a tariff of 100% on Country A's import of goods from Country B sector 2 at $T = 1500$.

enjoy similar growth whereas the country A suffers as it loses its comparative advantage. With the change in the focus on production from agriculture to manufacturing (or sector 1 to sector 2) in country B, country C has the advantage in the production of sector 1. While this result still satisfies [21] results, it also shows a scenario where a country could enjoy potential boosts in productivity, in the case that there is a productivity shock in another competing country and hence directing resources and attention away from the sector that the two countries competed in.

Appendix: Sectoral classification from EORA26 to 3-sector aggregation

Table A1: Sector Classification from EORA26 sectors into 3-sector system: primary, secondary, tertiary used for calibration.

Sector	Industries
Primary	Agriculture, Fishing, Mining and Quarrying, Food & Beverages, Textiles and Wearing Apparel, Wood and Paper, Petroleum, Chemical and Non-Metallic Mineral Products
Secondary	Metal Products, Electrical and Machinery, Transport Equipment, Other Manufacturing, Recycling, Electricity, Gas and Water, Construction
Tertiary	Maintenance and Repair, Wholesale Trade, Retail Trade, Hotels and Restaurants, Transport, Post and Telecommunications, Financial Intermediation and Business Activities, Public Administration, Education, Health and Other Services